低延迟视觉增强终端

面向低照度图像观察的 FPGA 像素流增强

中文版

PYNQ-Z2 · XC7Z020 · Vitis HLS 2026.1

1 背景与目标

问题

低照度图像噪声大、对比度低；软件增强一帧 3.1 s，无法连续观察。

思路

把整条增强链放进 PL 像素通路，像素进多少出多少，延迟以时钟计而不是以帧计。

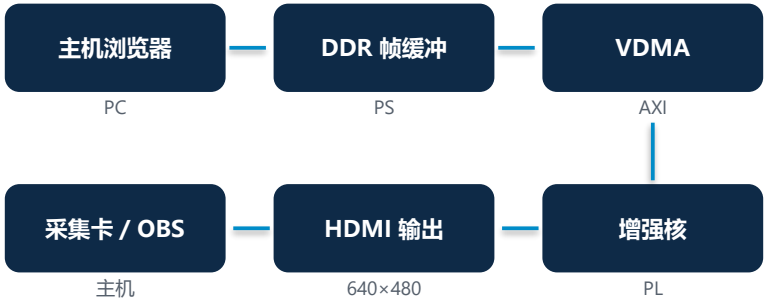
目标

不加帧缓存，同时做到画质可调、结果可复算、参数可在线切换。

方法

以逐位比对定正确性，以双时间戳定延迟，以寄存器写定切换。

2 系统架构



像素进多少出多少 • 链内无帧缓存

3 关键设计

三抽头行缓冲

每行缓冲每周期只读一次，II 保持为 1，8 个算子共用同一套读口。

参数存 BRAM

10 档预设写入寄存器即生效，不复位、不中断像素流。

全整数定点

与 NumPy 黄金模型 0 容差比对，8 档 2,764,800 通道逐位一致。

双时间戳

DMA 与 HDMI 两条通路各自打点，实测流水线延迟 16 拍 / 160 ns。

4 结果

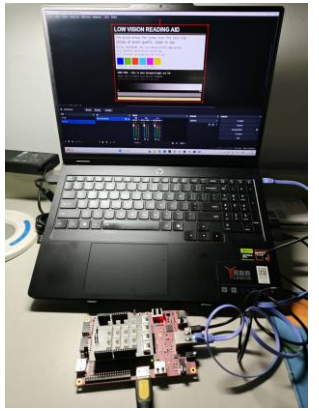
194.59 MHz
实现后频率，WNS +0.206 ns

1,659 LUT
占 XC7Z020 的 3%；FF 2,168 / BRAM 3 / DSP 2

15.6 ms
1280×720 单帧处理时间

196.7×
相对 ARM 纯软件的加速比（Sobel）

5 演示现场



照片：笔记本运行 OBS，经采集卡接收板卡 HDMI 输出。

复现：源码、构建脚本与黄金模型都在仓库，按 README 四步重建。　验证：8 档预设逐位 0 差异，双时间戳实测 16 / 21 拍。　边界：HDMI 固定 640×480@60，延迟指 PL 内部流水线。

Low-Latency Vision Enhancement Terminal

English

PYNQ-Z2 · XC7Z020 · Vitis HLS 2026.1

Pixel-stream enhancement on FPGA for low-light image observation

1 Background & Goal

Problem

Low-light images are noisy and flat; software enhancement takes 3.1 s per frame.

Approach

The chain runs inside the PL pixel path: pixels in, pixels out, latency in clocks, not frames.

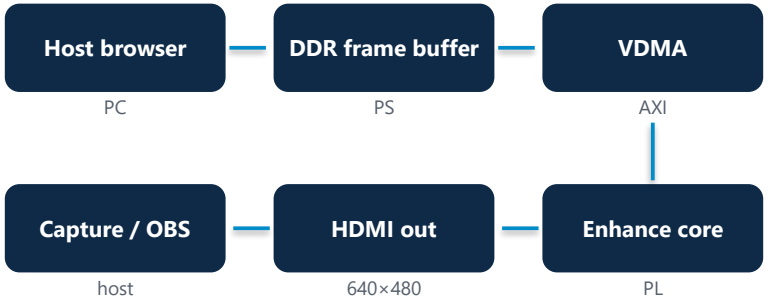
Goal

Adjustable quality, reproducible results and live preset switching, no extra frame buffer.

Method

Correctness by bit-exact compare, latency by dual timestamps, switching by one register write.

2 System Architecture



pixels in, pixels out · no frame buffer in the chain

3 Key Design

Three-tap line buffer

Each line buffer is read once per cycle, keeping II = 1 for all eight operators.

Parameters in BRAM

Ten presets take effect on a register write, without reset or stream interruption.

Integer datapath

Bit-exact against a NumPy golden model: 0 of 2,764,800 channels differ across 8 presets.

Dual timestamps

DMA and HDMI paths are timed separately; measured latency is 16 clocks / 160 ns.

4 Results

194.59 MHz
clock after implementation, WNS +0.206 ns

1,659 LUT
3% of XC7Z020; FF 2,168 / BRAM 3 / DSP 2

15.6 ms
per frame at 1280×720

196.7×
speed-up over ARM software (Sobel)

5 Demo setup

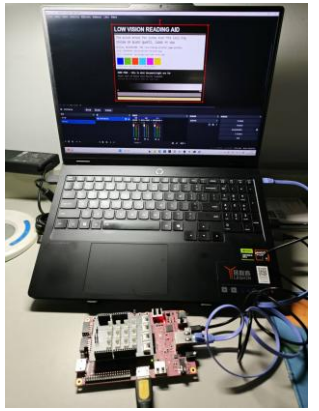


Photo: host running OBS, receiving the board's HDMI output through a capture card.

Reproduce: sources, scripts and golden model in the repo, four steps rebuild it. Verify: 0 differences across 8 presets, 16 / 21 clocks by dual timestamps. Scope: HDMI 640×480@60; latency is PL pipeline only.